InventionID: Automated Invention Extraction and Prior-Art Analysis from Technical Documents

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Abstract

Researchers and university technology-transfer teams often need to review technical papers to determine whether they contain potentially patentable ideas and whether similar prior art already exists. This process can be slow because papers are long, patent-search language differs from research-writing language, and reviewers need both an invention summary and supporting search evidence. In this project, we present InventionID, a web-based prototype that helps identify potentially patentable ideas from technical papers. Given a PDF document, the system extracts the likely invention described in the paper, generates patent-search queries, retrieves and analyzes related patents and academic papers, and compiles the findings into a structured report. This staged backend pipeline is exposed through an interface that demonstrates the end-to-end workflow: upload and progress tracking, visibility into the generated search queries, categorized result cards, and a structured JSON report. We test and evaluate this system on a number of examples and present results that highlight strengths as well as pointing to future directions for improvement. The project codebase is available at `https://github.com/VasistP/InventionID.git`.

1 Introduction

University technology-transfer offices and individual researchers regularly face the same question when a new technical paper is produced: does this work contain an invention worth protecting, and has something similar already been disclosed? Answering this question is the first step of an invention-disclosure review, and today it is almost entirely manual. A reviewer must read a long technical paper, mentally separate the patentable contribution from the surrounding scientific results, and then search patent databases for related prior art. Because research papers are written for scientific peers rather than for patent examiners, the language in the paper rarely matches the broader terminology used in patent claims, which makes the search step both slow and error prone. A reviewer can spend hours assembling a first-pass picture before deciding whether a disclosure deserves deeper, expert attention, and that cost means many disclosures are never reviewed thoroughly.

The goal of InventionID is to automate this first pass. The system takes an uploaded PDF, extracts a structured description of the likely invention, generates patent-search queries from that description, retrieves and analyzes related patents and academic papers, and compiles everything into a structured report. Rather than returning a single opaque answer, the system exposes each intermediate artifact, including the extracted invention, the generated queries, and the categorized results, so that a reviewer can inspect, adjust, and trust the output. We made this transparency a priority because the consequences of a patentability decision are significant, and a tool that shows its reasoning at every step is far more useful than one that hides it behind a single verdict. We want to be clear that InventionID is designed to *support* human review and to accelerate the early triage stage; it is not intended to replace expert or legal judgment about patentability. The contribution of the project is an end-to-end, transparent prototype that turns a technical PDF into a review-ready invention summary together with categorized prior-art evidence, and an evaluation of that prototype on representative

Preprint.

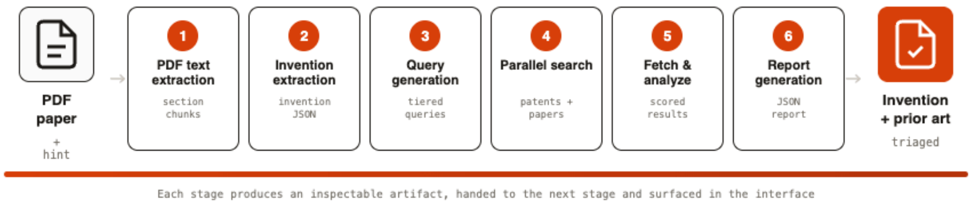

Figure 1: The InventionID pipeline. An uploaded PDF passes through six staged components; each stage produces an inspectable artifact that is handed to the next stage and surfaced in the interface, ending in a structured report of the extracted invention and categorized prior art.

papers. The remainder of this report gives the technical background for the problem, describes the system and each of its stages, presents our evaluation and results, lists individual contributions, and outlines next steps for a future team.

2 Background

A prior-art search establishes whether the key elements of a claimed invention have already been disclosed in earlier patents, patent applications, or published literature. In an invention-disclosure setting, the input is typically a technical paper describing a method or result, and the reviewer must make two related judgments: whether the paper contains a distinct, protectable invention rather than only a scientific finding, and whether closely related work already exists that would limit its novelty. Both judgments depend on first identifying what the invention actually is, meaning the specific combination of mechanism, components, and purpose that could be claimed, and then comparing it against the existing body of work.

The central difficulty is one of vocabulary and abstraction. A research paper describes a contribution in the precise language of its field, naming particular organisms, materials, reactions, or experimental conditions, and it foregrounds methods and results because those are what scientific peers evaluate. Patents, by contrast, describe inventions in broader, claim-oriented language designed to cover many variations of the same underlying idea, and they often use generic or legalistic phrasing in place of the specific terms a researcher would choose. As a consequence, a literal keyword search built from the paper's own wording frequently misses relevant patents that describe the same idea differently, while also returning loosely related noise. Effective prior-art review therefore requires bridging this gap: isolating the underlying invention, re-expressing it in search-appropriate terminology at several levels of generality, and then organizing the retrieved documents so that the strongest matches are easy to find and the reviewer's limited attention is spent where it matters most. These three needs, namely accurate extraction, vocabulary translation, and relevance-ordered presentation, directly motivate the design of InventionID.

3 Technical Approach and Methods

3.1 System Overview

InventionID is organized as a staged backend pipeline exposed through a web interface. A reviewer uploads a technical PDF, optionally accompanied by a short hint describing the suspected invention, and the system processes it through a sequence of well-defined stages, each of which produces an artifact that is handed to the next stage and surfaced in the interface. The main stages, shown in Figure 1, are: (1) PDF text extraction, (2) invention extraction, (3) search-query generation, (4) patent and academic-paper search, (5) result fetching and relevance analysis, and (6) final report generation.

We adopt a staged pipeline rather than a single large model call for three reasons. First, each stage can be validated and debugged independently, which makes the system far easier to build and reason about than one monolithic prompt whose failures are hard to localize. Second, every intermediate artifact can be shown to the reviewer, supporting the transparency and auditability that the application demands. Third, individual stages can be tuned, swapped, or re-run without disturbing the rest of the system, which matters because the extraction and search components have very different requirements and failure modes. The remainder of this section describes each stage in turn, followed by the user interface and output format.

3.2 PDF Text Extraction

The first stage converts the uploaded PDF into clean, section-aware text. Rather than treating the document as a single undifferentiated block, the extractor chunks the paper into sections so that downstream stages can reason over coherent units, such as the abstract, methods, and results, instead of arbitrary page breaks. This sectioned representation matters because the signal needed to identify an invention is concentrated in particular parts of a paper, and preserving that structure lets later stages weigh the relevant sections appropriately. The cleaned, sectioned text becomes the grounding context for invention extraction.

3.3 Invention Extraction

The invention-extraction stage is the analytical core of the system, and it is implemented as a multi-turn, self-correcting loop rather than a single prompt. In each iteration the language model extracts a structured description of the invention; that output is then validated against a required-field schema, and a patentability check scores the candidate against a three-facet rubric that considers the type of innovation described, whether the work is a concrete implementation rather than an abstract idea, and whether it has a practical, measurable application. The results of validation and the rubric are fed back into the conversation as observations, so the model sees its own errors and rubric feedback and can correct the extraction on the following turn instead of being re-prompted from scratch. This design lets the system recover from common failure modes such as missing fields, vague claims, or mischaracterized contributions within a single coherent interaction.

Two further design decisions make this loop practical and trustworthy. First, because the full document context would otherwise be resent on every turn, the document is cached and reused across iterations, which keeps the cost of repeated turns low even for long papers. Second, because a model's assessment of its own output can be systematically biased, the final confidence does not rely on self-assessment alone: it combines the model's own confidence with an independent score from a separate judge model that does not see the first model's self-rating. The loop continues to refine the invention until the blended confidence crosses a threshold or a maximum number of iterations is reached, at which point the structured invention, with its title, description, and key technical features, is passed downstream.

3.4 Search-Query Generation

Given the extracted invention, this stage produces the patent-search queries that drive prior-art retrieval. To bridge the gap between research and patent vocabulary identified in the background, it does not search with the paper's terms directly. Instead it first expands the invention's key concepts with synonyms and alternative phrasings, for example mapping a specific organism or method named in the paper to the broader terms a patent might use, and then generates a set of queries at several levels of specificity. Narrow queries target the exact mechanism and substrate to catch direct overlap, intermediate queries combine the mechanism with synonymous terms to catch near analogs, and broader queries reach for the wider claim family to catch contextually related patents the narrow queries would miss. Producing a tiered set of queries, rather than one query, increases the chance of surfacing relevant prior art across the different levels of abstraction at which it may be described.

3.5 Patent and Academic-Paper Search

The generated queries are executed against patent and academic literature sources. Because the query set is large and the individual searches are independent of one another, they are issued concurrently rather than sequentially, which keeps overall latency bounded even as the number of queries grows. Searching academic papers in addition to patents is deliberate, since closely related scientific work is itself a form of prior art and provides useful context for a novelty judgment. This stage returns the raw set of candidate patents and papers, which the next stage enriches and scores.

3.6 Result Fetching and Relevance Analysis

Raw search results contain only limited information, so for each promising candidate the system fetches the additional detail needed for assessment, such as the abstract, relevant dates, and assignee or author information. Each enriched result is then analyzed for relevance to the extracted invention and assigned both a relevance score and a category that reflects the strength of overlap. The three categories, *potentially blocking*, *relevant*, and *related*, are ordered by how directly a document bears on the invention, so that a reviewer can begin with the documents most likely to affect a patentability decision and work outward toward background context. This scoring and categorization step is what turns a flat list of search hits into an actionable triage.

3.7 Report Generation

The final stage merges the scored patents and papers, sorts them by category and relevance, attaches run metadata, and produces a structured JSON report capturing the invention summary, the generated queries, and the categorized prior art. This structured output serves two purposes: it is the persistent, machine-readable record of a run that could later be stored or exported, and it is the data that drives the web interface. Producing a single well-structured artifact at the end of the pipeline keeps the system's output consistent and easy to consume.

3.8 User Interface and Output Format

The web interface mirrors the pipeline so that the reviewer can follow the analysis from start to finish. It begins with an upload screen that accepts the PDF and an optional invention hint, followed by a live progress tracker that displays each stage as it runs so the reviewer can see the system working rather than waiting in front of a blank screen. Once extraction completes, the interface presents the structured invention summary, with its title, description, and key technical features, as shown in Figure 2, along with the generated search queries in a dedicated panel. Exposing the queries is important because patent results depend heavily on wording; showing exactly which queries were issued, together with how many results each returned (Figure 3), lets the reviewer understand why particular results appeared and adjust the search if needed. The retrieved patents and papers are then shown as result cards grouped under the labels *potentially blocking*, *relevant*, and *related* (Figure 4), ordered so that the highest-overlap documents appear first. The complete structured report is available as downloadable JSON, giving the reviewer both a readable on-screen view and a portable record of the analysis.

4 Results

4.1 Evaluation Plan

We evaluate InventionID along the three dimensions that correspond to the questions a reviewer cares about, using a combination of quantitative measurement and qualitative judgment on a set of example papers. For *invention-summary quality*, we read each extracted summary against its source paper and judge whether it captures the paper's actual contribution, includes the key technical features, and avoids introducing claims the paper does not support. For *search-query usefulness*, we inspect the generated queries and the number of results each returns, asking whether the queries are well targeted, neither so narrow that they return almost nothing nor so broad that they are uninformative, and whether they successfully translate the paper's terminology into patent-style phrasing. For the *relevance of returned prior art*, we manually review the documents the system places in the higher-priority categories and assess whether they are genuinely related to the invention, comparing them where possible against the prior art we would expect a human reviewer to find. Across all three dimensions we record quantitative measures, such as the number of queries generated, the number of patents and papers retrieved and analyzed, and the distribution of results across categories, and complement them with qualitative observations and informal feedback from project stakeholders.

4.2 Quantitative and Qualitative Results

On our example papers, the prototype reliably runs the complete workflow from PDF upload to final categorized report, confirming that the staged pipeline functions end to end without manual

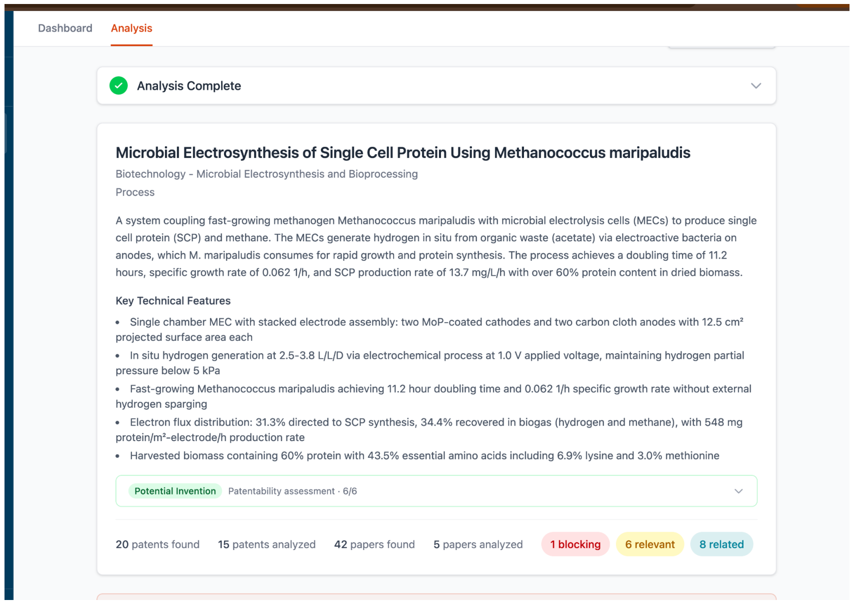

Figure 2: The invention-summary view. After extraction, the interface presents the invention title, domain, a plain-language description, and the key technical features, together with a patentability assessment and a summary of how many patents and papers were found and analyzed.

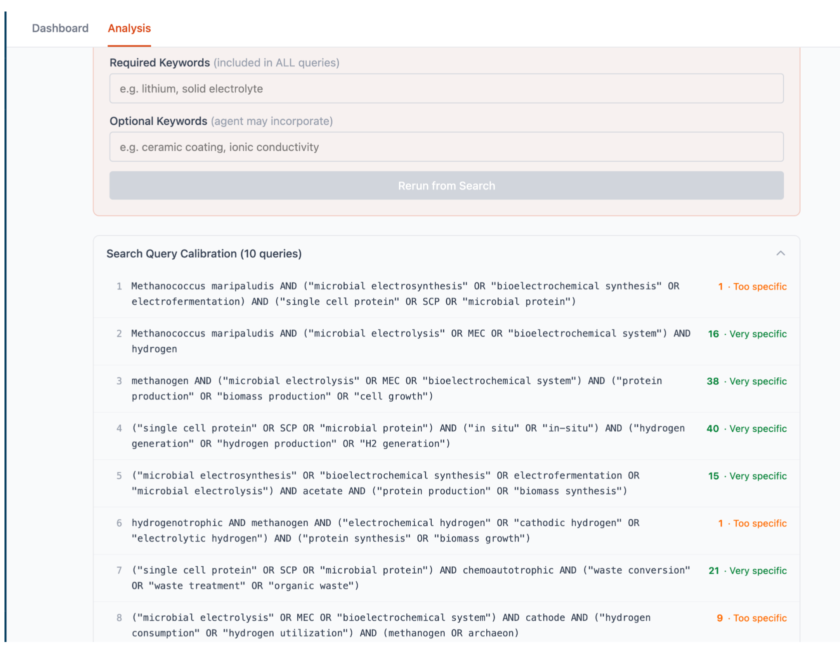

Figure 3: The search-query panel. The generated queries are shown along with the number of results each returns and a calibration label, making the search process transparent and letting the reviewer see which queries are well targeted and which are too narrow.

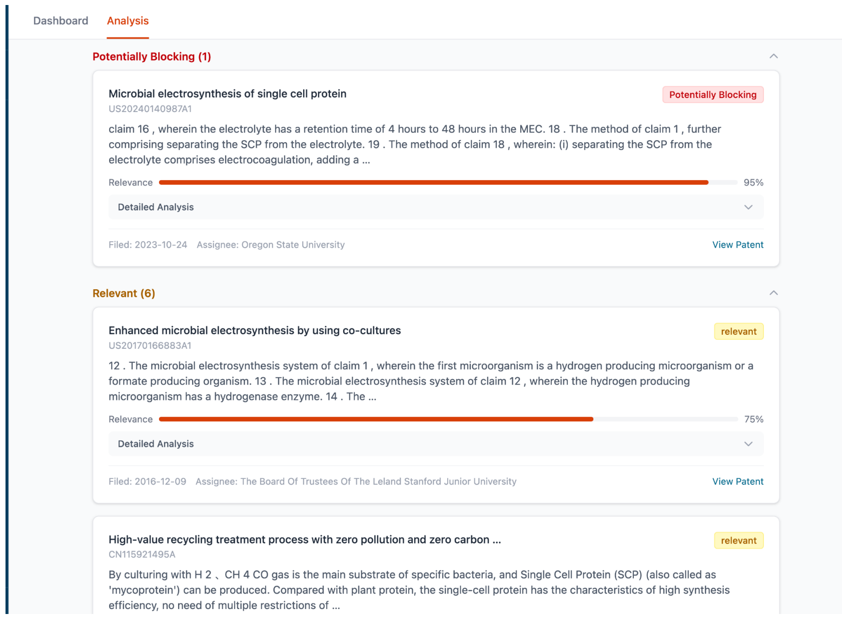

Figure 4: Categorized prior-art result cards. Retrieved patents are grouped as potentially blocking, relevant, or related, each with a relevance score, an excerpt, and source metadata, so the reviewer can inspect the highest-overlap documents first.

intervention. On a representative run that analyzed a paper on microbial electrosynthesis of single-cell protein, the system generated ten tiered queries, retrieved roughly 20 patents and 42 academic papers, and carried the most promising subset, 15 patents and 5 papers, through the full relevance analysis. The analyzed prior art was then distributed across the three categories: one document was flagged as potentially blocking, six as relevant, and the remainder as related background. This ordering let the reviewer inspect the highest-overlap documents first and treat the long tail of related results as optional context, which is the triage behavior the system is intended to provide. Table 1 summarizes the representative run.

Table 1: Representative end-to-end run on an example paper.

Measure	Value
Search queries generated	10 (tiered)
Patents retrieved	$\sim$20
Patents fully analyzed	15
Academic papers retrieved	$\sim$42
Academic papers fully analyzed	5
Potentially blocking / relevant / related	1 / 6 / 8

Qualitatively, the query-generation stage successfully reformulated paper-specific terminology into broader, patent-style phrasing, and the visible-query panel made it easy to see why particular results appeared and to identify queries that were too narrow to be useful, since these occasionally returned only a single hit. The extracted invention summaries were generally faithful to the source papers and surfaced the key technical features in a form a reviewer could quickly audit, which is the most important property for a first-pass tool. We also observed limitations that directly motivate our next steps. Results from the same patent family sometimes appear as separate entries, inflating the apparent number of distinct matches; the boundary between the *relevant* and *related* categories can be sensitive to wording and would benefit from clearer, explainable criteria; and search precision still depends noticeably on the quality of the generated queries. Overall, the results show that InventionID already delivers a useful first-pass triage, reducing what is normally hours of manual searching to a single automated run whose every step is inspectable, while leaving clear, concrete room to improve search precision and result organization before the tool is ready for routine use.

5 Contributions

5.1 Pranav Vasist

Pranav Vasist led the majority of the system, spanning both the web interface and significant portions of the backend. On the front end, he designed and implemented the upload screen and invention-hint input, the live progress tracker that surfaces each pipeline stage as it runs, the invention-summary view, the search-query panel, and the categorized result cards. On the backend, he contributed to the staged pipeline and its integration, including the search components and the flow of staged outputs through to the interface, ensuring the end-to-end system operates as a coherent product rather than a set of disconnected components. He was also responsible for the structured on-screen presentation of results and the downloadable report view.

5.2 Pranav Pravin Mane

Pranav Mane focused on document processing and parts of the extraction logic. He implemented the PDF text-extraction and section-chunking stage that produces the section-aware grounding context for the rest of the pipeline. He also worked on the multi-turn ReAct extraction loop, in particular the mechanism for persisting and carrying forward the data produced across iterations, and contributed to result validation by checking the system's outputs against expectations on example papers.

5.3 Prayoga

Prayoga led patent-search support and the prompt-engineering work behind the system's analytical stages. This included the search-query generation and the patent and academic-paper retrieval components, along with the parallel execution of searches across sources. Prayoga designed the three-facet patentability rubric, covering invention type, patent eligibility, and practical application, used to judge candidate inventions, and crafted the prompts that drive invention extraction and relevance analysis. Prayoga also drove testing with example documents and the preparation of evaluation materials, including assembling the example papers used to exercise the system and comparing returned prior art against expectations.

6 Next Steps

6.1 System Improvements

The most important remaining work concerns the quality and presentation of search results. We plan to improve query refinement and patent-result ranking so that the most relevant documents rise reliably to the top, and to add clearer, explainable criteria for why each result is categorized as potentially blocking, relevant, or related. This will both make the triage more trustworthy and address the category-boundary sensitivity we observed in evaluation. We also plan to group results that belong to the same patent family so that duplicate entries no longer crowd the result list and inflate the apparent number of distinct matches. Finally, before the website is used more broadly, we will add secure deployment support such as HTTPS/SSL so that uploaded documents and generated reports are handled safely.

6.2 Documentation and Codebase

To make the project maintainable and extensible by a future student, the GitHub repository at `https://github.com/VasistP/InventionID.git` includes a README that gives an overview of the project, instructions for installing dependencies and running the system end to end, and a guide to extending individual pipeline stages. The README also describes the major components of the codebase, namely the document-processing stage, the extraction and analysis pipeline, the search components, and the web interface, so that the relationship between this report's description and the code is easy to follow. Our intent is for a new student to be able to clone the repository, run an example, and locate the part of the code they need to modify without additional guidance.

7 Conclusion

InventionID is an end-to-end prototype that converts an uploaded technical PDF into an invention summary, a set of patent-search queries, and categorized prior-art results for human review. By isolating the invention with a self-correcting extraction loop, translating it into search-appropriate language through tiered query generation, and organizing the retrieved patents and papers by relevance, the system addresses the three obstacles of accurate extraction, vocabulary translation, and relevance-ordered presentation that make first-pass invention review slow and tedious. Our evaluation on example papers shows that the prototype already supports a useful first-pass triage while keeping every step inspectable, and it points to concrete directions for improvement: better result ranking and de-duplication, clearer and explainable category criteria, and a hardened, secure deployment. Pursuing these directions would move InventionID from a working prototype toward a tool that technology-transfer reviewers could adopt in practice.